Environmental Science & Technology Letters

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Letter

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Per- and Polyfluoroalkyl Substances in Reusable Feminine Hygiene Products

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Cite This: *Environ. Sci. Technol. Lett.* 2025, 12, 924−929

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ABSTRACT: Personal care products, such as cosmetics, have received attention for containing per- and polyfluoroalkyl substances (PFAS), yet limited information is known on the PFAS content in reusable feminine hygiene products, both in the United States and abroad. In this study, 59 reusable feminine hygiene products across five product categories first underwent total fluorine screening via particle-induced gamma-ray emission spectroscopy. Period underwear and reusable pads had the greatest rates of intentional PFAS use (>110 ppm total fluorine) at 33% and 25%, respectively. Subsequently, 19 of these products were chosen for targeted analysis of 31 ionic and 11 neutral PFAS. PFAS were detected in 100% of the extracted products, with 6:2 and 8:2 fluorotelomer alcohols being the most abundant compounds. Tested products were sourced from multiple markets, including North America, South America and Europe, all of which had at least one item with levels consistent with intentional fluorination. The observation of at least one product in each tested category not containing intentional fluorine indicates that PFAS use is unnecessary in all these reusable products. The results of this small initial survey, together with limited knowledge of the dermal absorption of PFAS, suggest more comprehensive PFAS exposure studies are needed.



KEYWORDS: *Reusable feminine hygiene products, PFAS, period underwear, PIGE, fluorotelomer alcohols (FTOHs), fluorotelomer methacrylates (FTMAcrs)*

INTRODUCTION

Per-and polyfluoroalkyl substances (PFAS) are a class of over 14,000 anthropogenic compounds that provide appealing physical properties such as water and oil repellency, and stain resistance.[1,2] This, in turn, has led to the addition of PFAS to consumer products, such as nonstick cookware, food packaging, and textiles.[3−8] Personal care products, such as cosmetics,[9,10] sunscreen,[11] and dental floss,[12,13] have been investigated previously due to the potential for direct human exposure to PFAS from these products.

Feminine hygiene products are a group of single-use and reusable products, including tampons, sanitary pads, and menstrual cups, that are used during menstruation. One study by Zhou et al. investigated the use of PFAS in 91 Chinese personal hygiene products using liquid chromatography tandem mass spectrometry (LC-MS/MS) analysis. Products analyzed included sanitary pads, panty liners, tampons, paper diapers, menstrual cups, and bactericidal liquids. The products found to contain the highest sum of PFAS concentrations were paper diapers, sanitary pads, and menstrual cups, however these results are likely an underestimate of the true concentrations as only ionic PFAS were measured.[14] In addition to that study, intentional fluorination has also been detected in period underwear. A 2020 Sierra Magazine article described two brands of period underwear tested for their total fluorine (total F) concentrations. One of the brands returned over 3,000 ppm of F, indicating intentional fluorine use,[15] which led to a lawsuit against the brand that was settled in January 2023 for $5 million.[16] However, with many brands and types of feminine hygiene products available, much is still unknown about the occurrence of PFAS in these products, especially outside of China. Additionally, the widespread use of PFAS in the growing market of reusable feminine hygiene products has yet to be studied.

Feminine hygiene products are generally worn against the skin, posing a direct exposure pathway. However, the dermal absorption of PFAS has not widely been studied yet, and the few studies to date have investigated a limited number of PFAS through targeted analyses using LC-MS/MS. In one *in vitro* study, Franko et al. tested the dermal penetration of predominantly nonionized perfluorooctanoic acid (PFOA) in human skin, observing that 24% of the applied dose passed through the full-thickness skin and epidermis over 24 h. They also examined the pH dependency of dermal uptake, showing that skin penetration was greatly reduced when PFOA was in

Received: June 3, 2025
Revised: July 7, 2025
Accepted: July 8, 2025
Published: July 22, 2025



ACS Publications

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its ionized form, which more closely reflects typical environmental exposures.[17] In a separate study, Abraham and Monien assessed real-world dermal absorption by applying $^{13}C_4$−PFOA spiked sunscreen to a single human subject for 48 h. Plasma levels measured over the course of 115 days estimated that 1.6% of the PFOA dose was dermally absorbed into the blood plasmsa.[18] More recently, Ragnarsdóttir et al. used *in vitro* 3D human skin equivalent experiments to estimate the absorbed fraction of C5−C14 PFCAs and C4−C9 PFSAs. The study found that absorption rates decreased with increasing chain length, with PFPeA having the greatest absorbed PFAS at 58.9 ± 1.9% after a 36-h exposure.[19] Another study modeling the dermal absorption of neutral PFAS suggests that some neutral analytes, including *N*-methyl- and N-ethyl-perfluorobutyl sulfanomido ethanol (N-MeFBSE and N-EtFBSE) and *N*-methyl- and N-ethyl-perfluorohexyl sulfonamido ethanol (N-MeFHxSE and N-EtFHxSE), are significantly more likely to be absorbed through the skin than inhaled.[20] These initial studies suggest that dermal absorption may represent a significant exposure pathway for PFAS when present in feminine hygiene products worn against the skin. In addition to direct dermal absorption concerns, intentionally PFAS-treated feminine hygiene products will also contribute to end-of-life environmental pollution for these "forever" chemicals as a broader exposure concern.

This work presents an investigation into the occurrence of PFAS in numerous types of reusable feminine hygiene products, including period underwear, reusable pads, reusable incontinence underwear, and menstrual cups. Additionally, a sampling of sanitary pads, panty liners, tampons, and disposable incontinence underwear were analyzed to confirm the Zhou et al. results with other product markets. These single-use product findings are compiled in the Supporting Information. Most of the products analyzed were sold in North America, but a subset of products from South America, Europe, and Asia-Pacific were also examined. Particle-induced gamma-ray emission (PIGE) spectroscopy was used to screen 59 reusable feminine hygiene products for the presence of fluorine. Previously PIGE has been used to measure total fluorine in several other consumer products, including textiles,[8,21] cosmetics,[10] and food packaging.[5] Samples then underwent targeted PFAS analysis of 31 analytes which were measured using liquid chromatography tandem mass spectrometry (LC-MS/MS) and an additional 11 analytes which were measured using gas chromatography−mass spectrometry (GC-MS). Through these analyses, this study aims to identify trends in fluorination across different feminine hygiene product classes and markets.

METHODS AND MATERIALS

Sample Acquisition. A total of 59 new reusable feminine hygiene products were acquired for analysis through purchase or by donation, and consisted of numerous brands from several countries (Tables S1 and S2). Products consisted of 43 period underwear, 8 reusable pads, 4 menstrual cups, 3 reusable incontinence underwear, and 1 reusable incontinence pad.

PIGE Sample Preparation and Analysis. All 59 reusable products underwent total fluorine screening. Products with multiple layers, such as period underwear and reusable pads, had each individual layer screened. In total, 323 unique samples underwent total fluorine screening via PIGE. In preparation for PIGE, samples were cut into 2.5 cm × 1.5 cm pieces and mounted on metal target frames for ion beam analysis at the St. Andre Facility within the Notre Dame Nuclear Science Laboratory.[22] Samples were bombarded by a 3.95 MeV beam of protons with a current of ∼ 75 nA *ex-vacuo* for 3 min each. As inorganic fluoride is assumed to be negligible in textiles, F signals were inferred to correspond only to organic fluorine (i.e., PFAS).[23] Paper standards consisting of NaF solutions (0−750 ppm) spiked and dried on Whatman 1 filter paper (Sigma-Aldrich, St. Louis, MO) were run daily with the samples to allow for interday normalization with the NaF spiked fabric standards. As PIGE is a surface technique, typically penetrating 100−200 μm into a textile, sample thickness was measured for each sample and used to choose the appropriate fabric calibration curve to convert total fluorine signals into concentrations as described in Xia et al.[8]

LC-MS/MS Sample Preparation. A subset of samples was selected for targeted analysis to ensure representation from each market and most product classes, prioritizing those with high total F based on PIGE results, along with a smaller number determined to contain no intentional F. A subset of 19 products (14 period underwear, 3 reusable pads, 2 reusable incontinence underwear) resulting in 59 unique samples were prepared for targeted analysis using a methanol extraction. Briefly, a 2 cm × 2 cm swatch was cut of each sample, spiked with 1 ng of isotopically labeled internal standards (Wellington, Guelph, ON) and added to 3 mL of LC-MS grade methanol (Fisher, Waltham, MA). The sample was sonicated for 30 min before the extracted solution was transferred to a new vial. The process was repeated with another 3 mL of methanol, resulting in 6 mL of final extract. At the conclusion of the extraction method, samples were concentrated to dryness under nitrogen, resuspended in 1 mL of 80:20 methanol−water mixture, and filtered with a 0.45 μm PES syringe filter (VWR, Visalia, CA). Then 250 μL was transferred to a polypropylene HPLC vial (Agilent, Santa Clara, CA) for LC-MS/MS analysis. A complete list of samples that underwent targeted analysis is found in the Supporting Information Excel file.

GC-MS Sample Preparation. Samples were also extracted once more for the targeted analysis of 11 neutral PFAS. The extraction method was a modification of a previously published method,[5] and is detailed in the Supporting Information. Briefly, new cuts of samples were sonicated for 30 min twice in 4:1 hexane:isopropanol and 1:1 methanol:acetonitrile. Samples that were known to contain polyfluoroalkyl phosphate esters (PAPs) from prior LC-MS/MS analysis were then cleaned up using a 200-mg, 3 mL Strong-Anion Exchange (SAX) cartridge (Phenomenex, California).

LC-MS/MS and GC-MS Analyses. A targeted analysis method for 31 ionic PFAS analytes was performed on an Agilent 1290 Infinity II UHPLC system coupled to an Agilent 6470B triple quadrupole mass spectrometer. A targeted analysis method for 11 neutral PFAS analytes was performed on a Hewlett-Packard 6890 GC coupled with a Hewlett-Packard 5973 MSD. A complete list of measured analytes, internal reference standard pairings, chromatographic and ion source parameters, complete dMRM parameters, limits of detection and quantification for each analyte, and the complete QA/QC information are included in the Supporting Information (Tables S3−S11).

RESULTS AND DISCUSSION

Total Fluorine Screening Using PIGE. A total of 59 reusable products were analyzed with PIGE. Figure 1 shows

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Environ. Sci. Technol. Lett. 2025, 12, 924−929

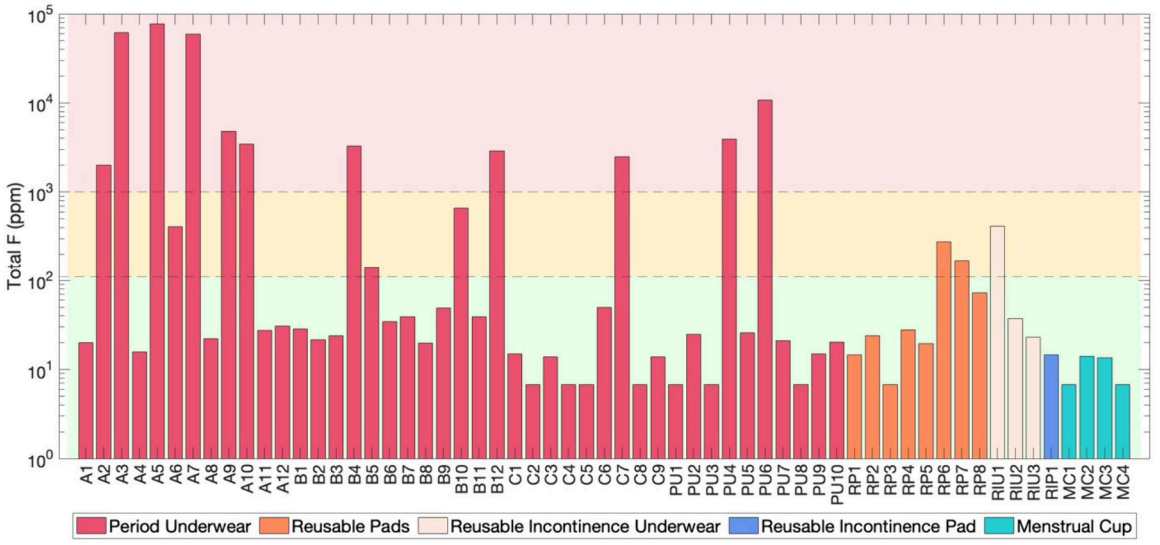


Figure 1. Distribution of log-scaled ppm total F measured using PIGE for each reusable feminine hygiene product category. Individual bars are color coded by product type. Products with multiple layers are reported as the result of the single layer containing the greatest fluorination. The background colors correspond to the three fluorine concentration categories: no intentional fluorination (<110 ppm, green), nonpolymeric PFAS (110−1000 ppm, yellow), and polymeric PFAS (>1000 ppm, red).

the breakdown of fluorination by product category. A distribution of the PIGE results by product category are displayed in Figure S1. Concentrations of total fluorine were characterized into three different categories: no intentional fluorination (<110 ppm, 71.2% of samples), intentional fluorination via nonpolymeric PFAS (110−1000 ppm, 10.2%), and intentional fluorination via polymeric fluorine (>1000 ppm, 18.6%). The lower limit of 110 ppm for the nonpolymeric PFAS category was set using the "middle value" presented for the functional concentration range of nonpolymeric PFAS in school uniforms.[8] This is similar to the 100 ppm limits for total F in textiles set by OEKO-TEX[24] and legislation passed in several US states.[25−27] Nonpolymeric PFAS are assumed to be present through the use of some spray-on textile impregnation agents.[3,28] The 1000 ppm lower limit for polymeric PFAS was determined from an industrial recommendation for fluoropolymer concentrations on fabrics.[29]

The highest fluorine concentrations were observed in some of the period underwear samples, in which 11 of the 43 (26%) tested samples contained at least 1000 ppm of F and concentrations up to 77,000 ppm were observed. In addition to period underwear, intentional fluorination (>110 ppm F) was observed in at least one of the tested reusable pads and reusable incontinence underwear. Additionally, all product categories contained at least one product without intentional fluorine concentrations.

The individual fabric layers in the reusable products were also analyzed. The number of layers present in these samples ranged from 3 to 8, with an average of 4 layers. Figure S2 displays the PIGE results for each layer. Often, intentional fluorination is only applied to a single layer in a product, but from this analysis there does not appear to be an industry standard for which layer is fluorinated.

Targeted Analysis Using GC-MS and LC-MS/MS. All extracted reusable products (period underwear, reusable pads, reusable incontinence underwear) had at least one detected PFAS and the sum of 42 targeted PFAS ($\sum_{42}$PFAS) concentrations ranged from 21.4 to 2200 ng/g, with a median 380 ng/g (Table 1, Table ES2). Neutral PFAS contributed the greatest percentage to the sum of targeted PFAS (98.4%).

Specifically, FTOHs and FTMAcrs dominated the PFAS measured, with detection frequencies of 100% and 47.4%. Concentrations of FTOH and FTMAcrs comprised 73.2% and 23.8% of the $\sum_{42}$PFAS, respectively. The abundance of 6:2

Table 1. Detection Frequencies (DF), Median and Range of Concentration (ng/g), and Contributions of the Most Abundant PFAS in Reusable Feminine Hygiene Product Samples (N = 19)[a]

	DF	Median (ng/g)	Range (ng/g)	% of Σ PFAS
PFCAs				
PFPeA	31.6%	0.032	ND−24.9	0.754%
PFHxA	26.3%	0.079	ND−5.87	0.207%
PFHpA	21.1%	0.031	ND−1.18	0.028%
PFOA	15.8%	0.055	ND−11.4	0.217%
PFTDA	15.8%	0.008	ND−1.12	0.019%
PFSAs				
PFHxS	10.5%	0.015	ND−1.40	0.016%
PAPs				
6:2 diPAP	10.5%	0.010	ND−1.28	0.014%
Neutrals				
6:2 FTOH	68.4%	84.9	ND−1320	47.8%
8:2 FTOH	94.7%	50.8	ND−621	18.1%
10:2 FTOH	47.4%	0.554	ND−252	7.24%
6:2 FTMAcr	47.4%	0.643	ND−1060	23.8%
Σ Ionic PFAS	78.9%	3.19	ND−42.7	1.6%
Σ FTOH	100%	254	21.4−1320	73.2%
Σ FTAcr	10.5%	1.55	ND−134	1.4%
Σ FTMAcr	47.4%	1.25	ND−1060	23.8%
Σ Neutral	100%	377	21.4−2180	98.4%
Σ PFAS	100%	380	21.4−2200	100%

[a]PFAS included in this table were detected in >10% in one of the three product categories. The full table is provided in the Supporting Information Excel file. Nondetect (ND) values were replaced by 1/2 MDLs and < LoQ values for the ionic PFAS were replaced with 2*MDLs for median calculations.

https://doi.org/10.1021/acs.estlett.5c00553
Environ. Sci. Technol. Lett. 2025, 12, 924−929

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FTOH, 8:2 FTOH, 10:2 FTOH, and 6:2 FTMAcr suggests that these products may contain side-chain fluorinated polymers (SFPs).[30] SFPs can degrade over time, releasing perfluoroalkyl acids and contributing significantly to environmental PFAS contamination.[7,31] These values are consistent to the findings of PFAS in other textile studies, which have reported 85.93−99.96% of the ∑PFAS were neutrals.[8,32−34]

Placing these results in the context of the dermal absorption risk for the wearer, the most abundant ionic analyte in these products, PFPeA, has been shown by Ragnasdóttir et al. to have nearly 60% dermal absorption in skin models.[19] Literature on the dermal absorption of neutral PFAS is limited to the dermal-to-inhalation ratios of 34 candidate compounds calculated in Kissel et al. Of the neutral PFAS observed in this study, only 10:2 FTOH and 6:2 FTMAcr met the criteria for inclusion in Kissel's candidate list, and both of which were found to be more likely to be inhaled.[20] However, the findings from Franko et al. that determined nonionic PFOA was much more likely to penetrate the dermis than ionic PFOA[17] may suggest that neutral PFAS would behave similarly to nonionic PFOA and pose a significant dermal absorption risk.

Comparisons between Total Fluorine and Targeted Results. The $\sum_{42}$PFAS was 2 to 5 magnitudes smaller than the total fluorine concentrations measured with PIGE for the reusable feminine hygiene products (Figure S4). This fluorine mass balance gap indicates the use of other fluorinated compounds, such as fluoropolymers, which are not detected directly in targeted methods. The gap observed here is consistent with other studies utilizing both total fluorine methods and targeted methodologies in consumer products.[8,9,34,35]

Samples classified as no intentional F via PIGE ranged in $\sum_{42}$PFAS from 21 to 880 ng/g (median: 75 ng/g), whereas samples categorized as containing intentional fluorination had $\sum_{42}$PFAS concentrations falling between 48 and 2205 ng/g (median 847 ng/g). No noticeable trends were observed between the PFAS profiles of samples characterized as containing polymeric PFAS vs nonpolymeric PFAS (Figure S5).

Market Differences. This sample set included similar period underwear products purchased across different markets including: North America (15), South America (12), Europe (13), and Asia-Pacific (3). Total F concentrations in the range of intentional fluorination were observed in 7 pairs of South American underwear, 4 North American pairs, and 2 European pairs (Figure S6). At least one underwear layer was extracted for targeted analysis from each market and targeted results showed that all extracted pairs of underwear contained PFAS (Figure S7), demonstrating PFAS were present in some products regardless of product market or origin. While the subset of samples chosen for targeted analysis was small, the South American, Asia-Pacific and European samples contained primarily 6:2 FTMAcr and 6:2 FTOH, while the majority of the PFAS signature in North American products was 8:2 FTOH.

IMPLICATIONS AND LIMITATIONS

Several possible explanations exist for the addition of PFAS to reusable feminine hygiene products, including to prevent leakage or transfer of fluids outside of the product, or to impart stain resistance to the textiles. Importantly, the observations that at least one product in each category did not contain intentional fluorination and that 71.2% of all reusable products had no intentional PFAS use clearly indicates that PFAS are not an essential component of any of these products. The lifespan of period underwear is 6 months to approximately 2 years,[36] after which the performance of the product may decline. In the case of PFAS-containing period underwear, this decline may be due to textile degradation that results in the loss of fibers containing SFPs, which can occur via weathering[7] or washing.[31] Additionally, laundering of PFAS-containing textiles has been observed to decrease PFAA concentrations and wash out neutral PFAS,[33] resulting in the release of PFAS to sewage systems. Furthermore, the contributions of period underwear to landfills at the end of their product lifetimes also represent a route for PFAS to be released to the environment.[37] Therefore, these products are likely contributing to indirect PFAS exposure in addition to direct exposure via dermal absorption.

Notably, the use of reusable feminine hygiene products, shown in this study to often contain intentional fluorination, are on the rise due to environmental concerns related to single-use products. A 2021 Australian study found that nearly 50% of participating women aged 15−29 had used a reusable menstrual product.[38] Recent legislation banning PFAS use in menstrual products has passed in several US states, including Minnesota, Vermont, Maine, California, Colorado, and Rhode Island.[27,39−43] While passage of these recent legislations is a promising start, this study highlights the need of more information on the chemical composition of feminine hygiene products and accordingly more regulations.

Further studies are especially needed to better define the dermal exposure risks of PFAS across a wider variety of feminine hygiene products. This study represents a small survey of products and markets, with a focus on period underwear and the US market, and therefore broader research is necessary to draw stronger conclusions on PFAS presence and trends in the feminine hygiene industry. Additionally, specific experiments on dermal exposure from these products are needed, especially regarding neutral PFAS. These experiments are beyond the scope of this study due to the complexities of different product types and different PFAS used, their varying contact with the body, and the contribution of different layers in reusable products. Ultimately, this study reports the first confirmation of PFAS used in reusable feminine hygiene products and calls attention to the potential for PFAS exposure that feminine hygiene products pose and highlights the need for future studies and regulations.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.estlett.5c00553.

Additional experimental and quality control details, as well as parameters used for the LC-MS/MS and GC-MS/MS analyses; list of feminine hygiene brands tested and their country of purchase; figures and table examining trends observed in samples (PDF)

Table listing extraction methodology for LC-MS/MS samples; full results of PIGE and targeted LC-MS/MS and GC-MS analyses (XLSX)

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Environ. Sci. Technol. Lett. 2025, 12, 924−929

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Notes

The authors declare no competing financial interest.

ACKNOWLEDGMENTS

We gratefully acknowledge Emily Ewell, as well as the *New York Times* Wirecutter for the contribution of product samples. We would also like to acknowledge the Indiana University Sustainability Research Development Grant for funding the analysis of neutral PFAS at Indiana University. We also gratefully acknowledge the Center for Environmental Science and Technology (CEST) at the University of Notre Dame for use of their TurboVap during sample preparation of LC-MS/MS extracts.

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Environ. Sci. Technol. Lett. 2025, 12, 924−929